

Notes: Dougal, Engelberg, García, Parsons (RFS 2012)

Journalists and the Stock Market

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1 What the paper is doing

1.1 Research question

Do financial journalists *move* aggregate stock returns, or do they merely *follow* market events?

1.2 Core identification idea

The paper exploits the *exogenous scheduling/rotation* of *Wall Street Journal* columnists who write the daily “Abreast of the Market” (AOTM) column. The key move is:

- treat the *identity of the columnist on day t* as quasi-random with respect to *future* market returns,
- then ask whether the DJIA excess return on day t and day $t+1$ depends on who wrote AOTM on day t .

Because AOTM is published at a regular time and columnist assignment is largely predetermined, the paper argues this variation supports a *causal* interpretation.

1.3 Two maintained assumptions (the “why causality is plausible” checklist)

The empirical design leans on:

1. **Persistent content differences across journalists.** Different authors exhibit stable stylistic / sentiment differences (even for similar underlying facts).
2. **No systematic selection on future returns.** Editors do not choose which journalist writes AOTM based on information about *tomorrow’s* market return.

Tables/Figures 1–2 and Tables 5–6 are essentially devoted to defending (2), while Table 7 and Figure 3 defend (1).

2 Data and variable construction

2.1 Unit of observation

Daily observations from 1970–2007:

- the AOTM columnist (author credit) each trading day,
- DJIA returns (plus dividend yield adjustment),
- controls: lagged returns, lagged volume, lagged volatility proxies, calendar dummies, etc.

2.2 Excess return on the DJIA (the paper’s return measure)

Let p_t be the level of the DJIA index at the close of day t . Let $r_{f,t+1}$ be the one-month T-bill rate (used as the daily risk-free proxy in the paper). Define the DJIA *excess* return from close t to close $t+1$ as:

$$r_{t+1} = \left(\frac{p_{t+1} - p_t}{p_t} + dp_{t+1} \right) - r_{f,t+1}. \quad (1)$$

The price-weighted dividend yield of DJIA constituents is:

$$dp_{t+1} = \frac{\sum_{i \in DJIA} d_{i,t+1}}{\sum_{i \in DJIA} p_{i,t}}. \quad (2)$$

Interpretation: The authors work with DJIA excess returns because AOTM disproportionately covers the DJIA-type blue-chip stocks, so the DJIA is a natural target for “aggregate media influence.”

2.3 Journalist indicators

Let J index the 25 main journalists in the sample (those with at least 50 articles; the remainder are excluded or treated as “no author”). Define an indicator:

$$Journalist_{i,t} \equiv \mathbf{1}\{\text{journalist } i \text{ is credited for AOTM on day } t\}.$$

Note: the regressions include indicators both for *the journalist writing on day t* and *the journalist who wrote on day $t - 1$* , because the paper focuses on a two-day response window (publication day and the day after).

3 Scheduling evidence: why the author assignment can be treated as quasi-exogenous

This part is not “the return result” yet; it is the foundation for a causal interpretation.

3.1 Table 1: tenure and rotation statistics

What Table 1 contains. For each journalist: years active, number of AOTM articles, number of writing “rotations,” average run length, and the distribution across weekdays.

How I read it.

- *Overlap in active periods* matters: multiple journalists write in the same years, so journalist effects are not confounded by long-run time trends alone.
- *Rotation structure* matters: the same journalists tend to alternate, which produces frequent within-year switches that help identification.
- *Weekday specialization* matters: some journalists disproportionately write on Mondays (or Fridays), which later motivates the schedule-prediction models (Table 5).

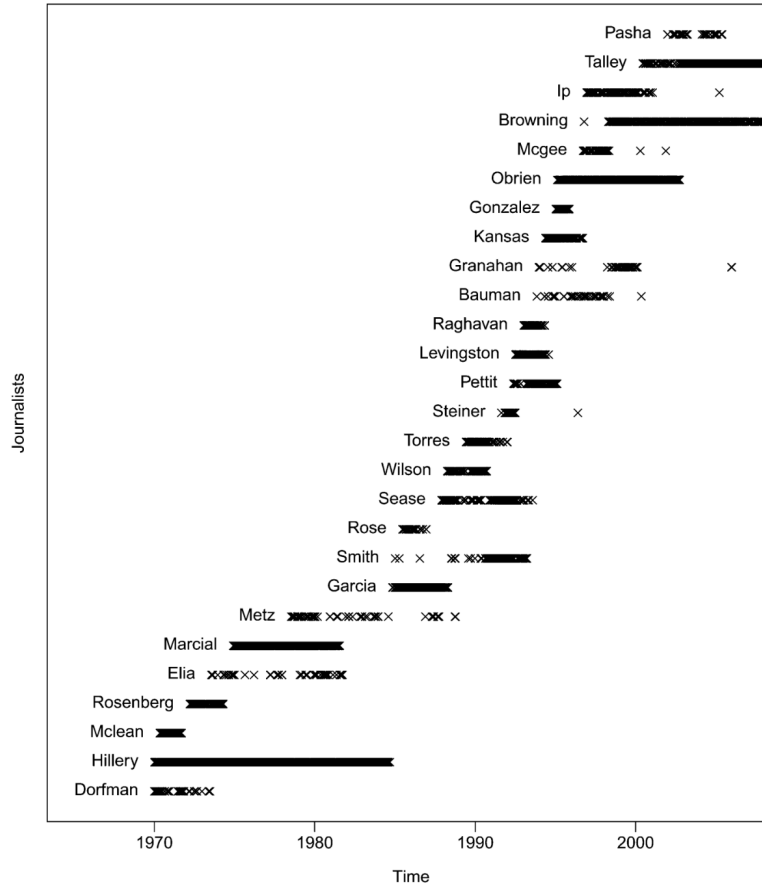


Figure 1
Timeline of AOTM journalists

3.2 Figure 1: full-sample timeline of authorship

What Figure 1 shows. A long timeline with one row per author and markers on the days they wrote. **Key point.** At any date there are *multiple* active authors; the pattern is not “one author per era.” This overlap is what allows journalist **fixed effects to be identified separately from time effects.**

3.3 Figure 2: three six-month schedule snapshots

What Figure 2 adds. Zoomed-in panels show that the schedule sometimes has clear regularities (e.g., one weekday writer plus a Monday specialist), but also that there is irregular turnover and substitutions (vacation coverage, sporadic contributors). **Why it matters.** Predictability of scheduling is not itself a problem for causality; what matters is whether that predictability is driven by *past schedule/calendar structure* rather than *future returns*. Tables 5–6 are devoted to this distinction.

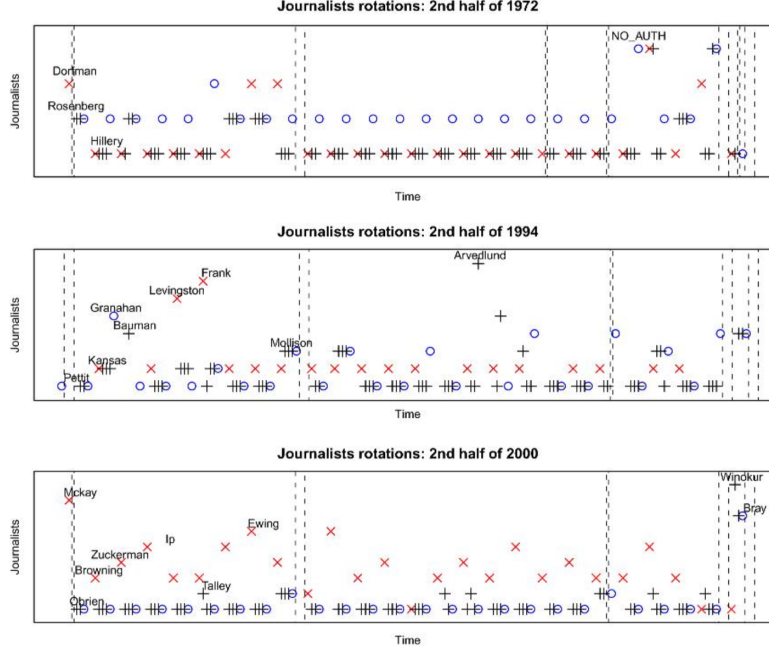


Figure 2
Sample of journalist writing days

4 Returns and journalists: baseline models and results

4.1 Table 2: univariate “mean return by journalist” comparisons

Table 2 reports, for each journalist:

- average DJIA excess return on the days the journalist writes ($\bar{r}^{wrote}$),
- average DJIA excess return on the day after the journalist writes ($\bar{r}^{dayafter}$),
- average DJIA excess return on all other days in that journalist’s tenure ($\bar{r}^{other}$),

with t-statistics.

Interpretation.

- The univariate evidence suggests heterogeneity: some journalists are followed by significantly positive or negative next-day returns.
- However, this is only a *first pass*: it does not control for return autocorrelation, volume/volatility clustering, or calendar patterns.

4.2 Table 3: multivariate return regression with journalist fixed effects (main model)

The baseline specification is:

$$r_t = c + \sum_{i=1}^J \left(\beta_{i,1} \text{Journalist}_{i,t-1} + \beta_{i,0} \text{Journalist}_{i,t} \right) + \eta' \text{Controls}_t + \varepsilon_t. \quad (3)$$

Table 2
Univariate return tests

	$\bar{r}_{\text{other}}$	$\bar{r}_{\text{wrote}}$	t -stat	$\bar{r}_{\text{day after}}$	t -stat
Hillery	-0.034	0.007	1.0	0.031	1.2
O'Brien	-0.036	-0.011	0.3	0.202***	2.6
Talley	-0.021	0.025	0.9	0.066	1.0
Marcial	0.046	0.036	-0.2	-0.055*	-1.8
Garcia	0.005	0.112	1.1	-0.249	-1.0
Smith	0.035	0.067	0.5	0.170	1.3
Wilson	0.107	-0.041*	-1.8	-0.020	-1.2
Browning	0.029	0.066	0.5	0.002	-0.4
Pettit	0.050	-0.057**	-2.0	0.084	0.5
Sease	0.026	0.053	0.3	0.133	1.2
Rosenberg	0.099	-0.095*	-1.9	-0.304***	-3.6
Kansas	0.076	0.119	0.6	-0.005	-1.0
McLean	0.103	0.069	-0.3	-0.035	-1.1
Raghavan	-0.024	0.200***	2.9	-0.021	0.0
Ip	0.013	0.195	1.4	0.118	0.8
Gonzalez	0.010	0.107	1.2	0.225**	2.3
Metz	0.029	0.031	0.0	0.041	0.1
Levingston	0.029	0.042	0.2	-0.031	-0.7
Pasha	0.011	0.275*	1.9	-0.431**	-2.3
Rose	0.123	0.034	-0.8	-0.084	-0.9
Steiner	0.057	-0.076	-1.6	0.142	0.7
Dorfman	0.027	0.028	0.0	-0.258***	-2.6
Bauman	0.073	0.001	-0.5	-0.064	-0.7
McGee	0.003	0.411**	2.4	0.406**	2.4
Elia	-0.005	-0.191	-1.3	0.027	0.2

Table 3
Multivariate return regressions

	Publication date		Day after	
	β_0	t -stat	β_1	t -stat
Hillery	0.038	0.6	-0.007	-0.1
O'Brien	-0.121*	-1.9	0.202***	3.2
Talley	0.010	0.2	0.039	0.6
Marcial	0.078	1.0	-0.105	-1.4
Garcia	0.204*	2.0	-0.118	-1.1
Smith	-0.026	-0.3	0.144*	1.7
Wilson	-0.032	-0.4	-0.133	-1.5
Browning	0.064	0.8	0.024	0.3
Pettit	-0.098	-1.0	0.111	1.1
Sease	0.010	0.1	0.072	0.8
Rosenberg	-0.047	-0.4	-0.304***	-2.6
Kansas	0.030	0.3	0.050	0.4
McLean	0.091	0.7	-0.026	-0.2
Raghavan	0.175	1.3	0.037	0.3
Ip	0.110	0.9	0.255**	2.1
Gonzalez	-0.076	-0.5	0.206	1.4
Metz	0.073	0.6	-0.081	-0.6
Levingston	0.038	0.3	-0.068	-0.5
Pasha	0.487***	3.4	-0.296**	-2.1
Rose	0.285	1.3	-0.323	-1.5
Steiner	-0.134	-0.9	0.027	0.2
Dorfman	0.050	0.3	-0.321**	-2.2
Bauman	-0.016	-0.1	-0.134	-0.8
McGee	0.386**	2.5	0.536***	3.4
Elia	-0.118	-0.8	0.060	0.4
Observations	9592			
R^2_{adjFE}	0.028			
R^2_{FE}	0.038			
$H_0 :$	$\beta_1 = 0$		$\beta_0 = 0$	
p -value/ F -stat OLS	0.000	2.4	0.011	1.8
p -value/ F -stat WHITE	0.000	2.4	0.042	1.8
p -value/ F -stat NW5	0.000	2.7	0.025	1.6
			$\beta_1 = \beta_0 = 0$	
			0.000	1.9
			0.000	1.9
			0.000	2.2

Controls (what is in $Controls_t$). The paper includes rich time-series controls:

- five lags of daily DJIA excess returns,
- five lags of detrended log NYSE volume,
- five lags of a detrended volatility proxy (squared residuals),
- day-of-week dummies, a January dummy, and year fixed effects.

How to interpret the coefficients.

- $\beta_{i,0}$: average effect on same-day return when journalist i publishes on day t (holding controls fixed).
- $\beta_{i,1}$: average effect on return *today* when journalist i published *yesterday* (day-after window).

What Table 3 is telling me.

- Several journalists have statistically significant $\beta_{i,0}$ and/or $\beta_{i,1}$.
- The joint F-tests strongly reject that all journalist coefficients are zero.
- Adding journalist fixed effects increases R^2 noticeably (even though daily returns are noisy), which is the paper’s first “macro” sign that author identity contains incremental predictive content.

5 Conditional effects: when do journalists matter the most?

The paper’s conceptual tension:

- If journalists mainly *interpret* events, their effects should be strongest when there is something big to interpret (large market moves).
- If investor attention is limited, persuasion might actually be harder during volatile news-heavy times.

5.1 Table 4: interacting journalist identity with lagged returns

To test state-dependence, the paper estimates:

$$r_t = c + \sum_{i=1}^J \left[\beta_{i,1} Journalist_{i,t-1} + \beta_{i,0} Journalist_{i,t} + \gamma_{i,1} (r_{t-2} \cdot Journalist_{i,t-1}) + \gamma_{i,0} (r_{t-1} \cdot Journalist_{i,t}) \right] + \eta' Controls_t + \varepsilon_t. \quad (4)$$

Table 4
Multivariate return regressions with interactions

	Publication date		Day after		Publication date		Day after	
	β_0	<i>t</i> -stat	β_1	<i>t</i> -stat	γ_1	<i>t</i> -stat	γ_1	<i>t</i> -stat
Hillery	0.036	0.6	-0.007	-0.1	-0.011	-0.3	-0.031	-0.9
O'Brien	-0.113*	-1.8	0.205***	3.3	-0.149***	-4.2	-0.077**	-2.2
Talley	0.017	0.3	0.042	0.6	-0.211***	-4.6	-0.014	-0.3
Marcial	0.072	0.9	-0.105	-1.4	0.089*	1.7	-0.103*	-1.9
Garcia	0.186*	1.8	-0.117	-1.1	-0.061	-1.4	0.048	1.1
Smith	-0.046	-0.5	0.162*	1.9	-0.090	-1.2	-0.095	-1.3
Wilson	-0.028	-0.3	-0.130	-1.5	-0.082	-1.1	-0.040	-0.5
Browning	0.071	0.9	0.025	0.3	-0.128*	-1.9	0.116*	1.7
Pettit	-0.107	-1.1	0.122	1.2	-0.004	0.0	0.058	0.5
Sease	-0.013	-0.1	0.091	1.0	0.056	0.8	-0.215***	-2.9
Rosenberg	-0.055	-0.5	-0.295**	-2.5	-0.025	-0.2	-0.136	-1.3
Kansas	0.030	0.3	0.027	0.2	-0.041	-0.3	0.039	0.2
McLean	0.096	0.7	-0.011	-0.1	0.319***	2.7	0.089	0.8
Raghavan	0.191	1.4	0.050	0.4	-0.170	-0.8	-0.018	-0.1
Ip	0.106	0.9	0.252**	2.1	0.039	0.4	-0.036	-0.4
Gonzalez	-0.092	-0.6	0.225*	1.6	0.004	0.0	-0.245	-1.2
Metz	0.072	0.5	-0.103	-0.8	-0.085	-0.7	0.063	0.5
Levingston	0.027	0.2	-0.057	-0.4	-0.056	-0.4	-0.021	-0.1
Pasha	0.487***	3.4	-0.283*	-2.0	-0.226**	-2.4	0.122	1.3

	Publication date		Day after		Publication date		Day after	
	β_0	<i>t</i> -stat	β_1	<i>t</i> -stat	γ_1	<i>t</i> -stat	γ_1	<i>t</i> -stat
Rose	0.297	1.4	-0.335	-1.5	-0.277*	-1.7	-0.037	-0.2
Steiner	-0.160	-1.0	0.057	0.4	-0.038	-0.2	-0.178	-0.9
Dorfman	0.042	0.3	-0.326**	-2.2	0.099	0.7	-0.182	-1.3
Bauman	-0.050	-0.3	-0.118	-0.7	0.252	1.3	-0.147	-0.8
McGee	0.352**	2.3	0.537***	3.4	-0.224**	-2.0	-0.200*	-1.8
Elia	-0.149	-1.0	0.050	0.3	0.146	1.0	0.073	0.5

Observations	9592
R^2	0.028
R^2_{adj}	0.050

$H_0 :$	$\beta_1 = \beta_0 = 0$	$\gamma_1 = \gamma_0 = 0$	$\beta_1 = \beta_0 = \gamma_1 = \gamma_0 = 0$
<i>p</i> -value/ <i>F</i> -stat OLS	0.000	1.9	0.000
<i>p</i> -value/ <i>F</i> -stat WHITE	0.000	2.2	0.000
<i>p</i> -value/ <i>F</i> -stat NW5	0.000	2.4	0.000

Interpretation of the interaction terms.

- $\gamma_{i,0}$ asks: when journalist i writes today, do today's returns depend more strongly on yesterday's return?
- $\gamma_{i,1}$ asks the analogous question for the day-after window: when journalist i wrote yesterday, do today's returns depend more strongly on the return from two days ago?

So γ 's are about whether a journalist *amplifies* or *attenuates* short-run return continuation (or reversal) after large moves.

What Table 4 implies.

- Many γ coefficients are significant for a meaningful subset of journalists.
- R^2 rises further relative to Table 3: the journalist effects are not purely constant shifts; they depend on the recent return environment.
- Joint tests reject that all interaction terms are zero, supporting the paper's claim that persuasion/interpretation is strongest during big market moves.

6 Endogeneity of journalist arrival: models and tests

A natural alternative story is editorial selection: an editor chooses certain writers for certain market environments. The paper tackles this directly.

Table 5
Linear probability models—forecasting journalists’ arrivals

	Model 1 R^2	Model 2 R^2	Model 3 R^2	$H_0 : \eta = 0$ p -value	$H_0 : \psi_i = \rho_i = \zeta_i = 0$ p -value
Hillery	0.038	0.062	0.389	0.592	0.000
O'Brien	0.028	0.088	0.429	0.434	0.000
Talley	0.287	0.318	0.618	0.847	0.000
Marcial	0.004	0.029	0.192	0.985	0.000
Garcia	0.103	0.118	0.476	0.735	0.000
Smith	0.273	0.281	0.508	0.965	0.000
Wilson	0.086	0.137	0.411	0.272	0.000
Browning	0.020	0.076	0.458	0.060	0.000
Pettit	0.072	0.117	0.320	0.675	0.000
Sease	0.081	0.125	0.446	0.002	0.000
Rosenberg	0.011	0.100	0.613	0.437	0.000
Kansas	0.073	0.174	0.462	0.028	0.000
McLean	0.023	0.111	0.586	0.355	0.000
Raghavan	0.050	0.081	0.174	0.855	0.001
Ip	0.085	0.102	0.350	0.990	0.000
Gonzalez	0.000	0.043	0.100	0.907	0.779
Metz	0.040	0.044	0.192	0.759	0.000
Levingston	0.122	0.176	0.349	0.864	0.000
Pasha	0.044	0.088	0.353	0.027	0.000
Rose	0.021	0.033	0.580	0.801	0.000
Steiner	0.224	0.241	0.406	0.010	0.000
Dorfman	0.008	0.029	0.355	0.465	0.000
Bauman	0.082	0.091	0.232	0.986	0.000
McGee	0.067	0.115	0.526	0.048	0.000
Elia	0.018	0.031	0.169	0.741	0.000

6.1 Table 5: linear probability models (LPMs) that predict who writes

For each journalist j , define the dependent variable $Journalist_{j,t}$.

The paper estimates three nested LPMs, reporting R^2 journalist-by-journalist:

Model 1 (year effects only).

$$Journalist_{j,t} = c_j + \sum_y \lambda_y Y_{y,t} + u_{j,t}, \quad (1)$$

where $Y_{y,t}$ are year dummies (capturing tenure timing).

Model 2 (add market variables).

$$Journalist_{j,t} = c_j + \sum_y \lambda_y Y_{y,t} + \theta' MarketVars_t + u_{j,t}. \quad (2)$$

Model 3 (add schedule persistence and calendar structure).

$$\begin{aligned} Journalist_{j,t} = & c_j + \sum_y \lambda_y Y_{y,t} + \theta' MarketVars_t \\ & + \sum_y \psi_y (Y_{y,t} \cdot Journalist_{j,t-1}) + \sum_y \rho_y (Y_{y,t} \cdot Journalist_{j,t-7}) + \sum_y \zeta'_y (Y_{y,t} \cdot D_t) + u_{j,t}, \end{aligned} \quad (3)$$

where $Journalist_{j,t-7}$ indicates whether journalist j wrote on the same weekday in the previous week, and D_t is the vector of day-of-week indicators.

How to interpret Table 5.

Table 6
Multivariate return regressions with IV

	Publication date		Day after		Publication date		Day after	
	β_0	t -stat	β_1	t -stat	γ_t	t -stat	γ_1	t -stat
Hillery	-0.033	-0.4	0.058	0.6	-0.047	-0.5	0.170*	1.7
O'Brien	-0.194	-1.3	0.118	0.9	-0.067	-0.6	0.066	0.6
Talley	0.061	0.5	0.007	0.1	-0.140	-1.1	0.028	0.2
Marcial	0.092	0.7	-0.037	-0.3	0.091	0.7	0.156	1.2
Garcia	0.251	1.4	-0.155	-0.9	-0.411	-1.5	0.648*	1.8
Smith	-0.222	-1.4	0.385***	2.6	0.067	0.4	0.154	0.9
Wilson	-0.156	-1.0	-0.066	-0.4	0.034	0.2	-0.026	-0.2
Browning	0.201	1.0	-0.079	-0.4	-0.193	-0.9	0.130	0.8
Pettit	-0.159	-1.2	0.164	1.1	0.057	0.3	-0.042	-0.2
Sease	-0.184	-0.7	0.294	1.4	0.314	1.4	-0.297	-1.4
Rosenberg	-0.089	-0.6	-0.287**	-2.0	-0.109	-0.7	0.457***	3.0
Kansas	-0.122	-0.8	0.069	0.5	0.029	0.1	0.193	0.9
McLean	0.007	0.0	0.195	1.1	0.258	1.6	0.213	1.0
Raghavan	0.288	1.0	0.052	0.2	0.063	0.1	0.035	0.1
Ip	-0.391	-1.2	0.196	0.7	0.537	1.6	-0.019	-0.1
Gonzalez	-0.259	-1.1	0.175	0.8	0.754*	1.8	-0.551	-1.5
Metz	-0.243	-0.7	0.062	0.2	0.021	0.0	-0.572	-1.4
Levingston	0.030	0.2	-0.113	-0.7	-0.186	-0.6	-0.043	-0.2
Pasha	0.291	0.9	-0.484	-1.3	-0.556**	-2.0	0.229	0.7
Rose	-0.183	-0.8	0.113	0.5	-0.736**	-2.5	0.569**	2.0
Steiner	-0.077	-0.3	0.129	0.5	0.372	1.1	-0.303	-0.8
Dorfman	0.029	0.1	-0.562***	-3.3	0.335	1.4	0.347**	2.2
Bauman	-0.547	-1.5	0.156	0.5	0.733*	1.8	0.412	1.0
McGee	0.358	1.1	0.712***	3.1	-0.509	-1.1	-0.601**	-2.1
Elia	-0.549	-1.1	-0.434	-1.1	0.588	1.1	-0.579*	-2.0
Observations	9592							
R^2_{JOJFE}	0.028							
R^2_{JFE}	0.054							
$H_0 :$	$\beta_1 = \beta_0 = 0$		$\gamma_1 = \gamma_0 = 0$		$\beta_1 = \beta_0 = \gamma_1 = \gamma_0 = 0$			
p -value/ F -stat BOOT	0.001	1.9	0.000	3.1	0.000	2.9		

- **Scheduling is predictable, but mostly not via market variables.** R^2 increases strongly when they include past writing patterns and weekday structure (Model 3), while tests for the market-variable block are generally weak.
- This supports the identifying assumption: journalist arrival is driven by *calendar/schedule mechanics*, not by market information about future returns.

6.2 Table 6: return regressions using predicted journalist assignment (IV-style)

Define $\widehat{Journalist}_{i,t}$ as the fitted value (predicted probability) from the schedule model in Table 5. The paper then reruns the return regression, replacing the *actual* journalist indicators by the *predicted* ones:

$$IVJournalist_{i,t} \equiv \widehat{Journalist}_{i,t}.$$

The regression mirrors (4), but with *IVJournalist* in place of *Journalist*.

Why this is persuasive. If the journalist effects were driven by endogenous editorial choice correlated with future returns, then stripping the regressors down to the *mechanical*, *predicted* schedule component should destroy the results. Table 6 shows the opposite: the qualitative patterns survive (though with reduced precision), and the joint significance remains strong.

Inference note. Because *IVJournalist* is a generated regressor, the paper uses bootstrap methods to compute t-statistics.

	Syllables		WPS		% Complex		Fog		Pessimism (Unconditional)		Pessimism (Interaction)	
	β_0	<i>t</i> -stat	β_0	<i>t</i> -stat	β_0	<i>t</i> -stat	β_0	<i>t</i> -stat	β_0	<i>t</i> -stat	β_0	<i>t</i> -stat
Hillery	0.002	0.5	-0.462***	-2.9	-0.092	-0.9	-0.223***	-3.0	-0.167***	-3.1	0.038	1.4
O'Brien	0.051***	15.5	-0.435***	-2.8	0.838***	8.2	0.164**	2.2	0.267***	5.1	-0.261***	-8.9
Talley	0.038***	11.0	-2.578***	-15.9	0.435***	4.1	-0.853***	-11.3	0.084	1.6	-0.265***	-7.0
Marcial	0.004	0.9	1.888***	9.6	0.079	0.6	0.785***	8.5	-0.293***	-4.5	0.051	1.2
Garcia	-0.006	-1.3	-0.164	-0.7	-0.411***	-2.8	-0.228**	-2.2	0.095	1.3	0.005	0.1
Smith	-0.017***	-3.9	-0.072	-0.3	-0.615***	-4.5	-0.276***	-2.9	0.297***	4.3	-0.110*	-1.8
Wilson	0.000	0.1	0.984***	4.8	-0.306*	-2.2	0.269***	2.8	-0.006	-0.1	-0.107*	-1.7
Browning	-0.082***	-19.3	2.667***	13.2	-2.203***	-16.5	0.189**	2.0	0.050	0.7	0.244***	4.2
Pettit	0.004	0.7	-1.023***	-4.2	0.161	1.0	-0.348***	-3.1	0.478***	5.9	-0.437***	-4.4
Sease	-0.025***	-5.0	0.857***	3.7	-0.353**	-2.3	0.200*	1.8	0.249***	3.2	0.097	1.6
Rosenberg	0.018***	2.9	0.694**	2.4	0.425**	2.2	0.447***	3.3	0.034	0.3	-0.033	-0.4
Kansas	0.029***	4.7	0.049	0.2	0.600***	3.1	0.254*	1.9	-0.076	-0.8	-0.230*	-1.8
McLean	0.008	1.1	-0.944***	-2.9	0.332	1.6	-0.249*	-1.7	-0.186*	-1.7	-0.248***	-2.6
Raghavan	-0.014*	-1.9	0.227	0.7	0.079	0.4	0.115	0.7	0.019	0.2	-0.284*	-1.7
Ip	0.009	1.3	0.994***	3.2	0.600***	3.0	0.643***	4.5	0.072	0.7	0.184**	2.3
Gonzalez	-0.001	-0.1	0.568	1.6	0.617***	2.6	0.471***	2.8	0.231**	2.0	-0.425**	-2.5
Metz	-0.006	-0.9	-0.091	-0.3	-0.435**	-2.1	-0.213	-1.4	-0.065	-0.6	0.032	0.3
Levingston	0.013*	1.8	-1.174***	-3.5	0.467**	2.1	-0.291*	-1.9	0.152	1.4	-0.144	-1.1
Pasha	0.003	0.4	-1.079***	-3.3	-0.166	-0.8	-0.494***	-3.2	0.261**	2.4	0.060	0.8
Rose	-0.006	-0.8	-0.598	-1.6	-0.404	-1.6	-0.395**	-2.2	-0.050	-0.4	-0.373***	-2.8
Steiner	0.015*	1.9	1.209***	3.3	0.882***	3.6	0.829***	4.8	-0.075	-0.6	-0.680***	-4.3

	Syllables		WPS		% Complex		Fog		Pessimism (Unconditional)		Pessimism (Interaction)	
	β_0	<i>t</i> -stat	β_0	<i>t</i> -stat	β_0	<i>t</i> -stat	β_0	<i>t</i> -stat	β_0	<i>t</i> -stat	β_0	<i>t</i> -stat
Dorfman	0.008	1.1	1.120***	3.0	0.379	1.6	0.599***	3.5	0.127	1.0	0.045	0.4
Bauman	0.029***	3.6	1.281***	3.4	0.932***	3.7	0.890***	5.0	-0.156	-1.2	-0.301*	-1.9
McGee	-0.015*	-1.8	1.994***	5.1	-0.340	-1.3	0.664***	3.6	-0.052	-0.4	0.086	0.9
Elia	0.004	0.5	-1.201***	-3.1	-0.158	-0.6	-0.548***	-3.0	0.556***	4.3	0.197	1.6
Observations	9552		9552		9552		9552		9552		9552	
R^2	0.199		0.391		0.190		0.319		0.295		0.295	
R^2_{JFE}	0.297		0.461		0.245		0.365		0.309		0.331	
p -value/ F -stat OLS	0.000	52.7	0.000	49.6	0.000	27.7	0.000	27.2	0.000	7.5	0.000	7.3
p -value/ F -stat WHITE	0.000	46.8	0.000	45.4	0.000	28.2	0.000	24.2	0.000	8.1	0.000	8.2
p -value/ F -stat NW5	0.000	40.3	0.000	39.5	0.000	26.1	0.000	21.3	0.000	7.5	0.000	7.5

7 What do journalist fixed effects “stand for”? Content-style evidence

Up to now, journalist identity is a reduced-form label. This section tries to connect it to measurable writing differences.

7.1 Table 7: article characteristics regressions

For each article characteristic (readability or tone), the paper estimates:

$$Characteristic_t = c + \sum_{i=1}^J \beta_i Journalist_{i,t} + \eta' Controls_t + \epsilon_t, \quad (4)$$

where $Characteristic \in \{\text{Syllables, WPS, \%Complex, Fog, Pessimism}\}$.

Definitions (what the variables mean).

- **Syllables:** average syllables per word.
- **WPS:** average words per sentence.
- **%Complex:** share of words with ≥ 3 syllables.
- **Fog:** Gunning Fog readability score (higher $\Rightarrow$ harder to read).
- **Pessimism:** %negative words – %positive words, normalized to mean 0 and variance 1.

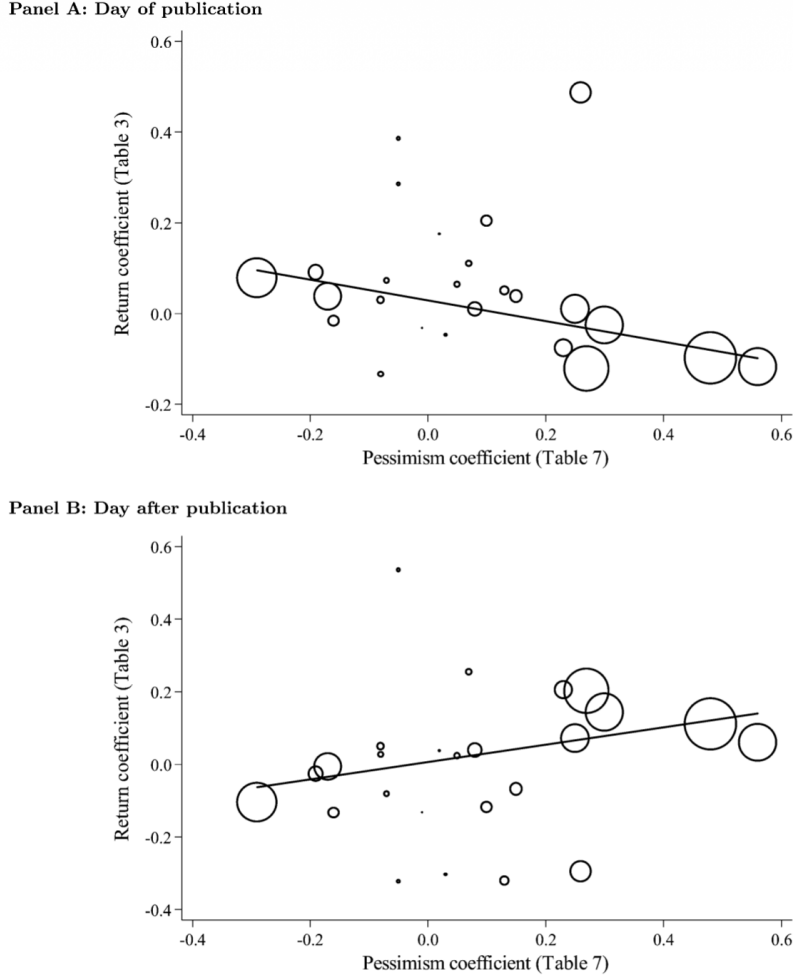


Figure 3
Linking journalist pessimism to returns

What Table 7 implies.

- Journalists have statistically distinct *styles* (readability and tone), consistent with persistent content differences.
- This is crucial: it makes it plausible that journalist fixed effects in return regressions proxy for real variation in interpretation/spin rather than pure noise.

7.2 Figure 3: linking measured pessimism to return effects

Figure 3 plots (weighted) relationships between:

- the journalist-specific pessimism coefficients from Table 7, and
- the journalist-specific return coefficients from Table 3 (publication day and day after).

How I read it. There is evidence that pessimistic tone is associated with more negative return responses for some journalists, but the paper is careful: automated word counts only capture a slice of what constitutes “content.”

Table 8 Multivariate return regressions and robustness								
	Open-Close	S&P 500	CRSP VWTD	GARCH-Adj.	Winsorized	Ten Authors	Calendar	Interactions
Panel A: $\beta_1 = \beta_0 = 0$								
<i>p</i> -value OLS	0.000	0.000	0.001	0.000	0.000	0.005	0.000	0.000
<i>p</i> -value WHITE	0.000	0.000	0.000	0.000	0.000	0.004	0.000	0.000
<i>p</i> -value NW5	0.000	0.000	0.000	0.000	0.000	0.002	0.000	0.000
<i>F</i> -statistic OLS	1.9	1.9	1.8	2.1	2.0	2.0	1.9	1.9
<i>F</i> -statistic WHITE	2.1	2.0	2.0	2.3	2.3	2.1	2.2	2.2
<i>F</i> -statistic NW5	2.3	2.2	2.2	2.5	2.5	2.2	2.3	2.2
Panel B: $\gamma_1 = \gamma_0 = 0$								
<i>p</i> -value OLS	0.000	0.000	0.000	0.004	0.000	0.000	0.000	0.000
<i>p</i> -value WHITE	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.012
<i>p</i> -value NW5	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.001
<i>F</i> -statistic OLS	2.4	3.3	3.8	1.6	2.4	4.0	2.3	3.8
<i>F</i> -statistic WHITE	2.1	2.9	3.0	2.0	2.5	2.8	2.1	1.5
<i>F</i> -statistic NW5	3.1	4.5	4.4	2.3	3.1	2.8	2.8	1.8
Panel C: $\beta_1 = \beta_0 = \gamma_1 = \gamma_0 = 0$								
<i>p</i> -value OLS	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>p</i> -value WHITE	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>p</i> -value NW5	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>F</i> -statistic OLS	2.2	2.6	2.8	1.9	2.2	3.0	2.1	2.9
<i>F</i> -statistic WHITE	2.2	2.5	2.6	2.1	2.4	2.4	2.1	1.8
<i>F</i> -statistic NW5	2.8	3.4	3.4	2.5	2.8	2.4	2.7	2.1

8 Robustness and specification checks

8.1 Table 8: robustness across return definitions and control sets

Table 8 repeats the Table 4 F-tests under alternative constructions:

- different return series (DJIA open-to-close, S&P 500, CRSP value-weighted),
- volatility-adjusted returns (GARCH),
- winsorized variables (reduce outlier influence),
- restricting to the top 10 most prolific journalists,
- adding calendar anomaly controls (holiday effects, turn-of-month, year-month FE),
- adding interactions that absorb time-varying return autocorrelation linked to volume/calendar.

Bottom line: p-values for joint tests of journalist coefficients and interaction terms remain very small across variants, suggesting the central result is not an artifact of a single specification choice.

8.2 Table 9: does “pessimism” explain journalist effects?

Table 9 uses word-count pessimism in two ways.

Panel A (add pessimism controls). Estimate:

$$r_t = c + \sum_{i=1}^J (\beta_{i,1} Journalist_{i,t-1} + \beta_{i,0} Journalist_{i,t}) + \zeta_1 Pessimism_{t-1} + \zeta_0 Pessimism_t + \eta' Controls_t + \varepsilon_t.$$

Interpretation: Even after adding *Pessimism*, journalist coefficients remain jointly significant. So journalist effects are not simply “the same as” measured word-count sentiment.

Table 9
Quantifying pessimism with word counts

	Panel A: Pessimism		Panel B: Interactions	
	Publication date	Day after	Day after	
	β_0	t -stat	β_1	t -stat
Hillery	0.037	0.6	-0.009	-0.1
O'Brien	-0.117*	-1.9	0.202***	3.2
Talley	0.010	0.2	0.041	0.6
Marcial	0.075	1.0	-0.107	-1.4
Garcia	0.205**	2.0	-0.119	-1.1
Smith	-0.021	-0.3	0.141*	1.7
Wilson	-0.031	-0.4	-0.135	-1.5
Browning	0.064	0.8	0.024	0.3
Pettit	-0.092	-0.9	0.110	1.1
Sease	0.015	0.2	0.068	0.7
Rosenberg	-0.045	-0.4	-0.307***	-2.6
Kansas	0.029	0.2	0.051	0.4
McLean	0.089	0.7	-0.029	-0.2
Raghavan	0.174	1.3	0.038	0.3
Ip	0.111	0.9	0.254**	2.1
Gonzalez	-0.074	-0.5	0.207	1.5
Metz	0.073	0.6	-0.084	-0.7
Levingston	0.040	0.3	-0.067	-0.5
Pasha	0.490***	3.4	-0.295**	-2.0
Rose	0.285	1.3	-0.324	-1.5
Steiner	-0.135	-0.9	0.029	0.2
Dorfman	0.052	0.4	-0.325**	-2.2
Bauman	-0.019	-0.1	-0.132	-0.8
McGee	0.383**	2.5	0.535***	3.4
Elia	-0.109	-0.7	0.056	0.4
	ζ_0	t -stat	ζ_1	t -stat
Pessimism	-0.014	-1.2	0.004	0.4
Observations	9592		9592	
R^2_{JOFE}	0.028		0.028	
R^2_{JFE}	0.038		0.041	
H ₀ :	$\beta_1 = \beta_0 = 0$		$\beta_1 = \beta_0 = 0$	
p -value/ F -stat OLS	0.000	1.9	0.000	1.8
p -value/ F -stat WHITE	0.000	2.0	0.000	1.9
p -value/ F -stat NW5	0.000	2.1	0.000	2.1

	2 days before		1 day before		Publication date		1 day after		Days after	
	$\beta_{i,-2}$	t -stat	$\beta_{i,-1}$	t -stat	$\beta_{i,0}$	t -stat	$\beta_{i,1}$	t -stat	$\beta_{i,2}$	t -stat
Hillery	0.113*	1.7	-0.028	-0.4	-0.005	-0.1	-0.057	-0.9	0.132**	2.0
O'Brien	0.005	0.1	0.122*	1.9	-0.130**	-2.0	0.229***	3.6	-0.026	-0.4
Talley	0.009	0.1	0.103	1.5	-0.013	-0.2	0.033	0.5	0.069	1.0
Marcial	0.101	1.3	0.019	0.2	0.033	0.4	-0.136*	-1.7	0.074	0.9
Garcia	0.039	0.4	0.020	0.2	0.226*	1.9	-0.058	-0.5	-0.142	-1.3
Smith	-0.171**	-2.0	0.114	1.3	-0.013	-0.1	0.129	1.5	-0.008	-0.1
Wilson	0.079	0.9	0.052	0.5	-0.064	-0.7	-0.118	-1.2	-0.070	-0.8
Browning	-0.043	-0.5	0.041	0.5	0.024	0.3	-0.012	-0.1	0.002	0.0
Pettit	-0.031	-0.3	0.015	0.1	-0.094	-0.9	0.105	1.0	0.031	0.3
Sease	-0.112	-1.2	0.096	1.0	0.027	0.3	0.068	0.7	0.044	0.5
Rosenberg	0.218*	1.8	0.035	0.3	-0.113	-0.9	-0.355***	-3.0	0.168	1.4
Kansas	-0.026	-0.2	0.180	1.5	0.008	0.1	0.045	0.4	-0.073	-0.6
McLean	0.209	1.6	-0.039	-0.3	0.032	0.2	-0.068	-0.5	0.058	0.4
Raghavan	-0.162	-1.2	0.146	1.1	0.177	1.3	0.046	0.3	0.023	0.2
Ip	-0.172	-1.4	-0.018	-0.1	0.088	0.7	0.248**	2.0	-0.323***	-2.6
Observations	9592									
R^2_{adj}	0.028									
R^2_{FEE}	0.044									
p -value/ F -stat OLS	0.814	0.7	0.842	0.7	0.077	1.4	0.001	2.2	0.672	0.9
p -value/ F -stat WHITE	0.671	0.9	0.583	0.9	0.153	1.3	0.000	2.4	0.541	0.9
p -value/ F -stat NW5	0.659	0.9	0.519	1.0	0.115	1.3	0.000	2.6	0.580	0.9

Panel B (journalist-specific pessimism slopes). Estimate:

$$r_t = c + \sum_{i=1}^J (\beta_{i,1} \text{Journalist}_{i,t-1} + \beta_{i,0} \text{Journalist}_{i,t}) + \sum_{j=1}^J \xi_{j,0} (\text{Journalist}_{j,t} \cdot \text{Pessimism}_t) + \eta' \text{Controls}_t + \varepsilon_t.$$

Interpretation: Allowing the pessimism effect to vary by journalist provides only limited incremental explanatory power. The paper takes this as evidence that “content” is hard to fully parametrize with simple word counts.

8.3 Table 10: falsification and event-window discipline

The paper augments (3) with additional lags and *leads* of journalist identity:

$$r_t = c + \sum_{i=1}^J \left(\beta_{i,2} \text{Journalist}_{i,t-2} + \beta_{i,1} \text{Journalist}_{i,t-1} + \beta_{i,0} \text{Journalist}_{i,t} \right) + \sum_{i=1}^J \left(\beta_{i,-1} \text{Journalist}_{i,t+1} + \beta_{i,-2} \text{Journalist}_{i,t+2} \right) + \eta' \text{Controls}_t + \varepsilon_t.$$

How I read Table 10.

- Effects are concentrated on publication day and the day after.
- Coefficients on *future* journalist assignments (leads) are not jointly significant, which is a clean falsification: “articles written for future publication” should not move today’s return.

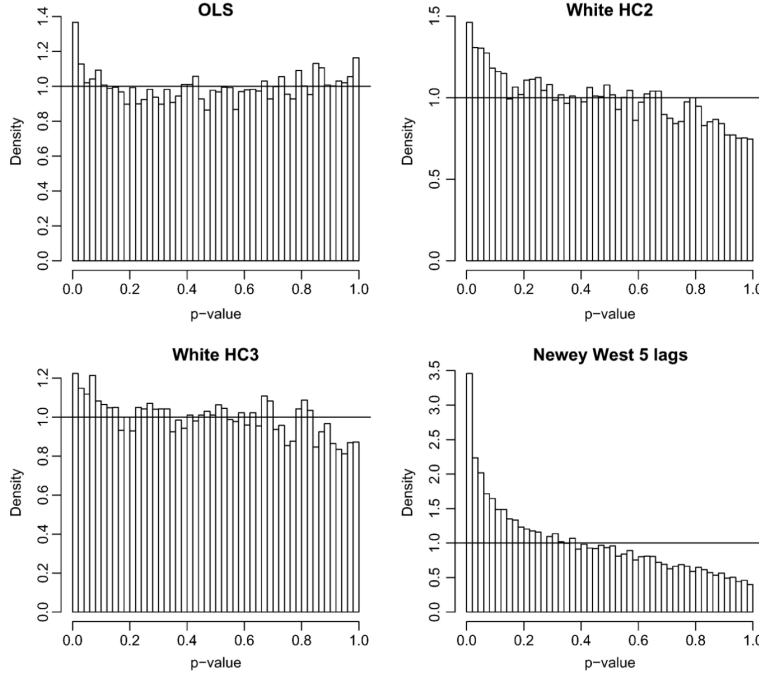


Figure 4
Histograms of p -values for the test $\beta_0 = \beta_1 = 0$.

8.4 Figure 4: finite-sample p -values for the key F-test

Figure 4 uses simulation/bootstrapping under the null (no journalist effects) to check whether the asymptotic p -values behave like $\text{Uniform}(0,1)$. **Interpretation:** Standard OLS inference can be mildly distorted in finite samples; robust and Newey–West procedures behave better. This motivates the paper’s emphasis on robust inference for the joint journalist tests.

9 Takeaways

1. **Causal reduced form:** journalist identity predicts DJIA returns over a short window, using scheduling variation that is plausibly exogenous to future returns.
2. **State dependence:** effects are stronger after large market moves; some journalists amplify and others dampen short-run serial correlation.
3. **Not driven by assignment endogeneity:** market variables do not forecast journalist assignment (Table 5), and predicted assignment still generates return effects (Table 6).
4. **Content is real but hard to parametrize:** journalists have distinct styles (Table 7), pessimism correlates with return effects (Figure 3), but simple word counts do not subsume journalist identity (Table 9).